



(12) **United States Patent**
Struik

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(54) **METHOD AND APPARATUS FOR PROVIDING AN ADAPTABLE SECURITY LEVEL IN AN ELECTRONIC COMMUNICATION**

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This patent is subject to a terminal disclaimer.

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H04L 9/40 (2022.01)

(52) **U.S. Cl.**
CPC **H04L 63/105** (2013.01); **H04L 63/12** (2013.01); **H04L 63/1441** (2013.01); **H04L 63/20** (2013.01); **H04L 63/0227** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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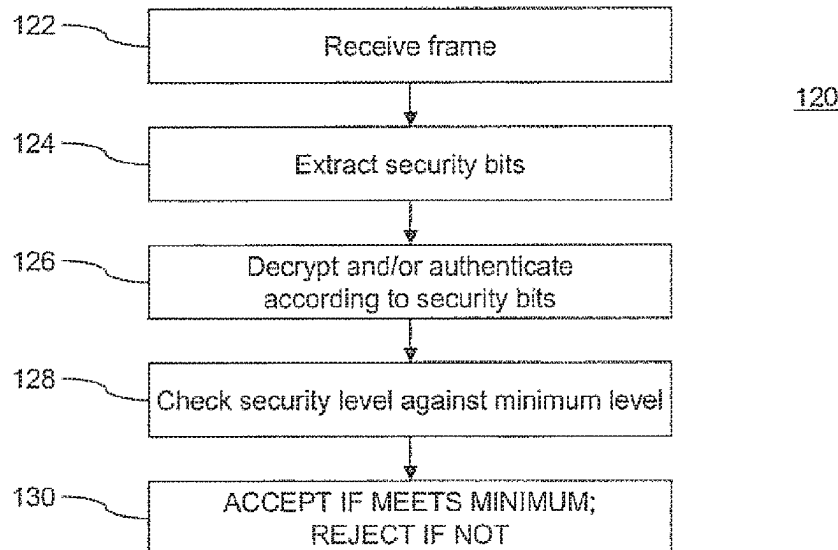
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(57) **ABSTRACT**

A method of communicating in a secure communication system, comprises the steps of assembling a message at a sender, then determining a frame type, and including an indication of the frame type in a header of the message. The message is then sent to a recipient and the frame type used to perform a policy check.

9 Claims, 8 Drawing Sheets



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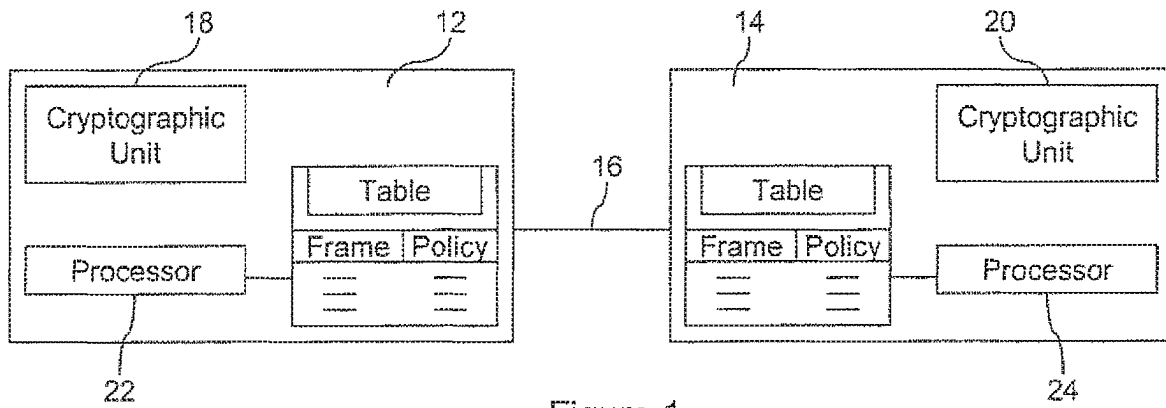


Figure 1

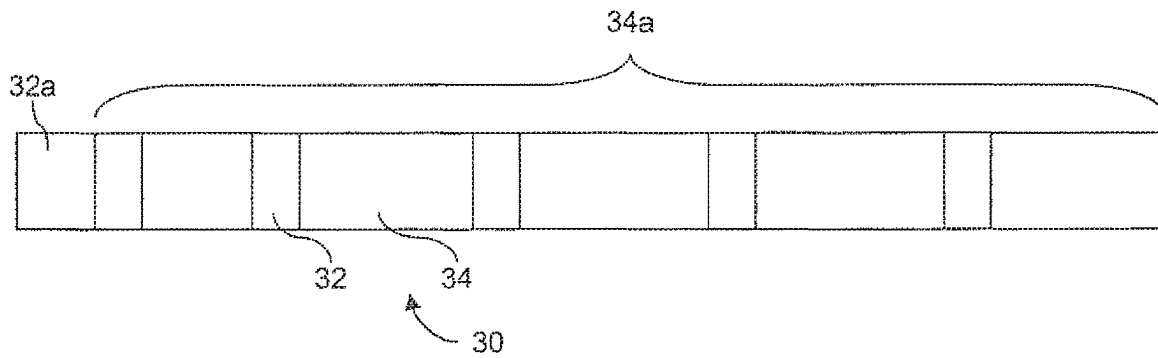


Figure 2

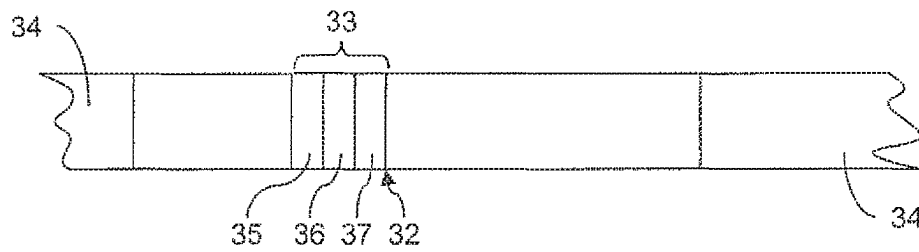


Figure 3

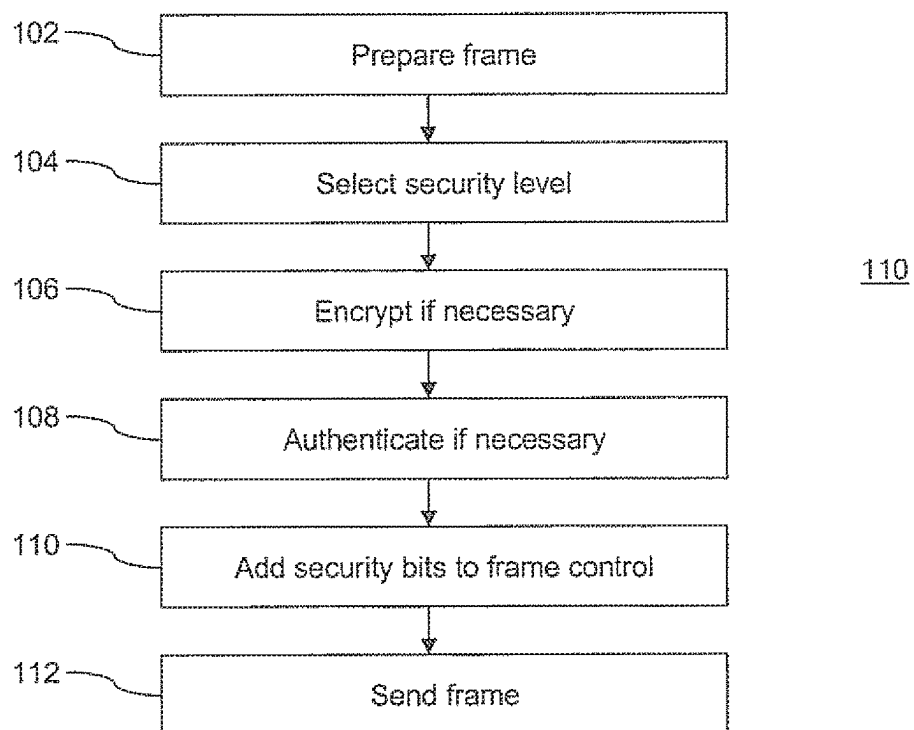


Figure 4

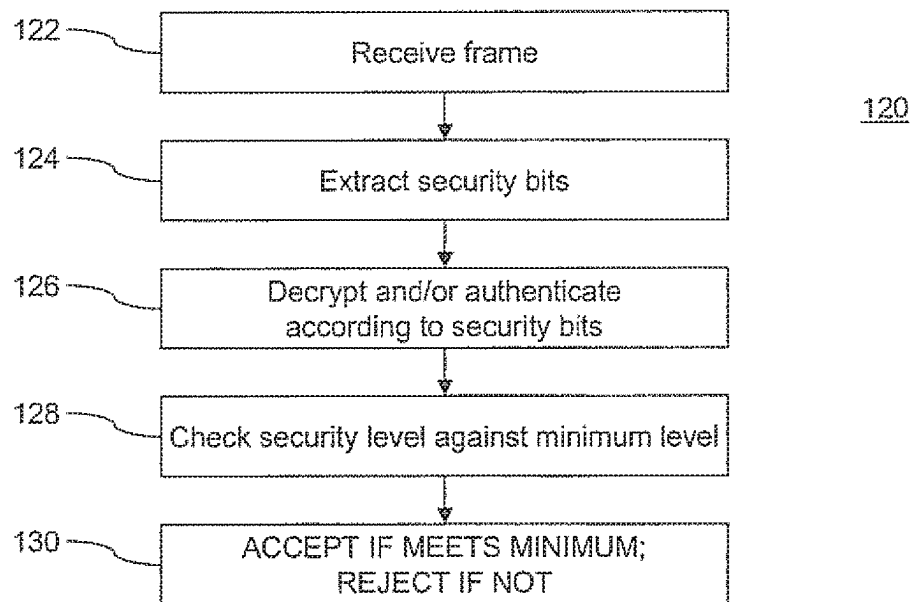


Figure 5

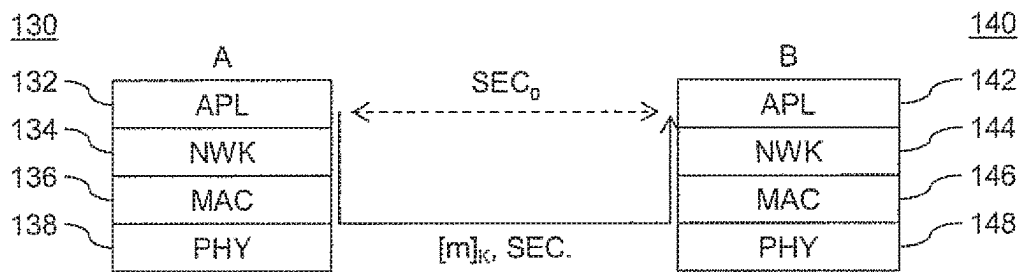


Figure 6

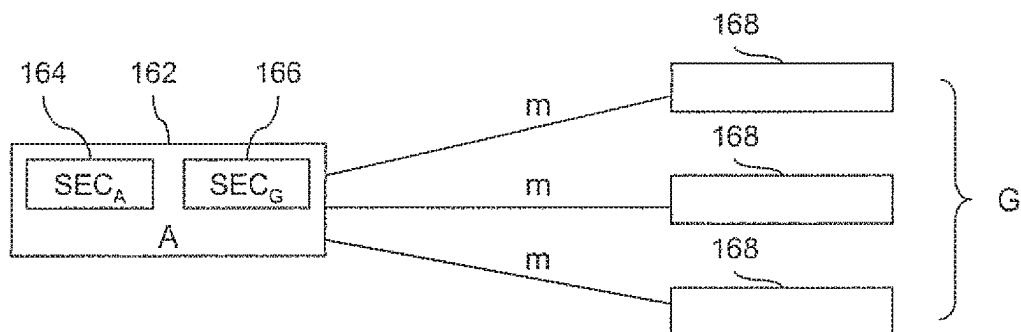


Figure 7

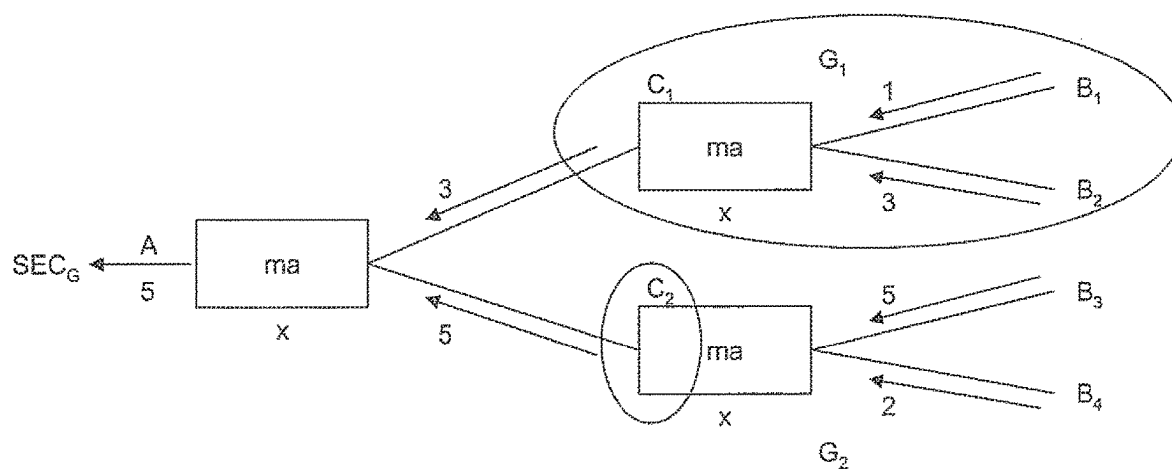


Figure 8

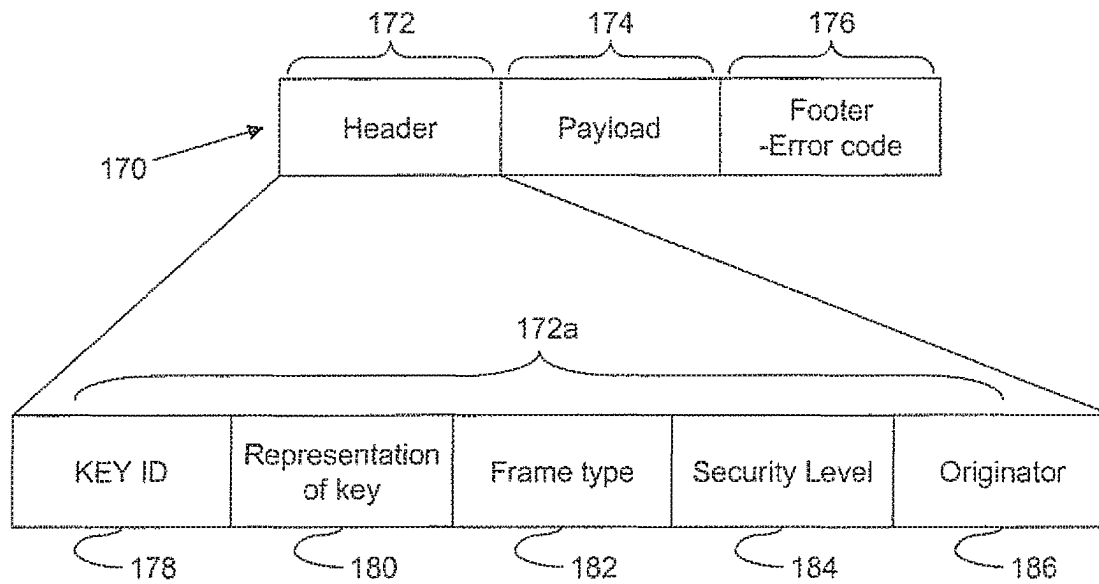


Figure 9

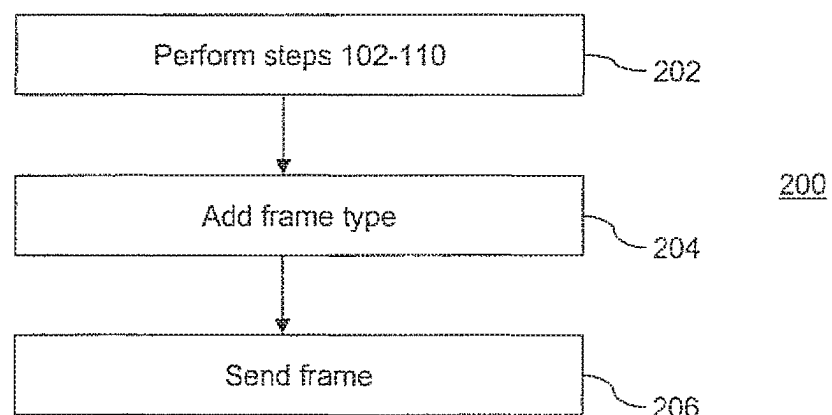


Figure 10

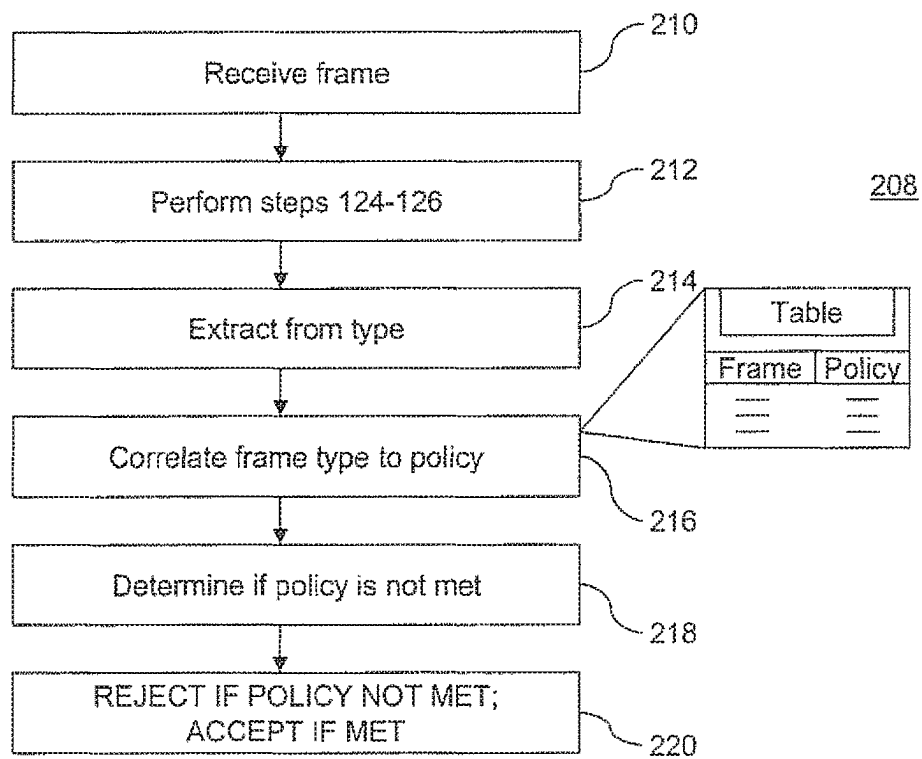


Figure 11

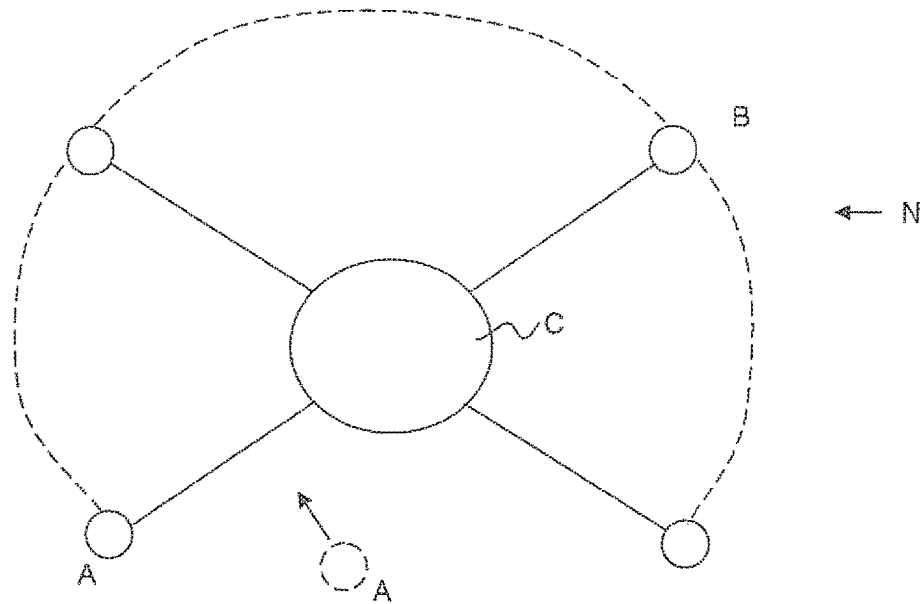


Figure 12

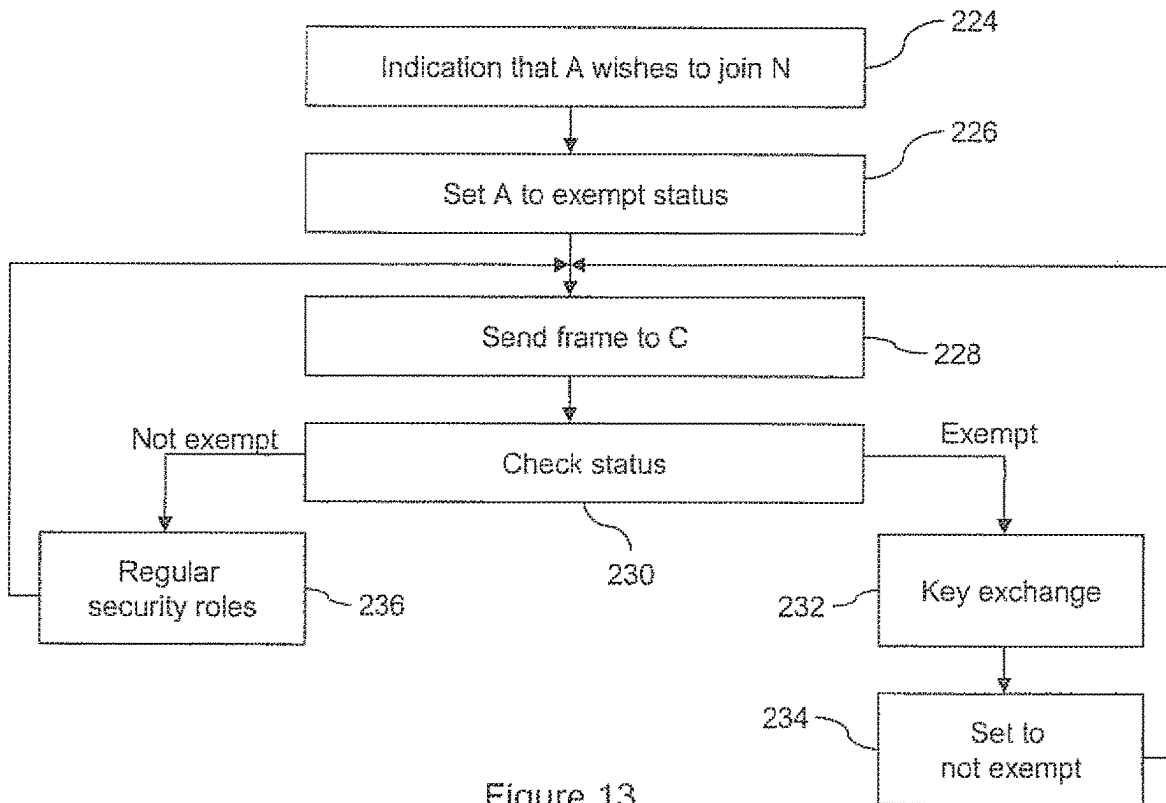


Figure 13

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METHOD AND APPARATUS FOR PROVIDING AN ADAPTABLE SECURITY LEVEL IN AN ELECTRONIC COMMUNICATION

This application is a continuation of and claims priority to U.S. patent application Ser. No. 16/153,676 filed on Oct. 5, 2018, which is a continuation of and claims priority to U.S. patent application Ser. No. 15/595,542 filed on May 15, 2017, which is a continuation of U.S. patent application Ser. No. 14/170,213 filed on Jan. 31, 2014, which is a continuation of U.S. patent application Ser. No. 11/735,055 filed on Apr. 13, 2007, now U.S. Pat. No. 8,688,978 which claims priority from U.S. Patent Application No. 60/791,434 filed on Apr. 13, 2006. The entire contents of the priority documents are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to a method and apparatus for providing an adaptable security level in an electronic communication.

DESCRIPTION OF THE PRIOR ART

In electronic communications, it is often necessary to prevent an eavesdropper from intercepting a message. It is also desirable to have an indication of the authenticity of a message, that is a verifiable identification of the sender. These goals are usually achieved through the use of cryptography. Private key cryptography requires sharing a secret key prior to initiating communications. Public key cryptography is generally preferred as it does not require such a shared secret key. Instead, each correspondent has a key pair including a private key and a public key. The public key may be provided by any convenient means, and does not need to be kept secret.

There are many variations in cryptographic algorithms, and various parameters that determine the precise implementation. In standards for wireless communications, it has been customary to set these parameters in advance for each frame type. However, this approach limits the flexibility of the parameters.

When one device is communicating with several other devices, it will often need to establish separate parameters for each communication.

It is an object of the present invention to obviate or mitigate the above disadvantages.

SUMMARY OF THE INVENTION

In one aspect, there is provided a method of communicating between a first correspondent and a second correspondent in a data communication system comprising assembling a data stream at said first correspondent, said data stream having at least one frame, said frame having a header and data; incorporating in said header, an indication of a frame type; and forwarding said frame to said second correspondent to enable said second correspondent to determine the acceptability of said frame according to said frame type.

In another aspect, there is provided a method of verifying a communication between a first correspondent and a second correspondent in a data communication system comprising said second correspondent: receiving from said first correspondent, a frame having a header and data, said header including an indication of a frame type; determining said

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frame type from said header; and correlating said frame type to a policy to determine if said frame type is acceptable for at least one attribute of said frame.

In yet another aspect, there is provided a method of communicating between a pair of correspondents in a data communication system comprising exempting one of said pair of correspondents from security rules associated with said communication system to enable said one correspondent to initialize communication with the other of said correspondents.

BRIEF DESCRIPTION OF THE DRAWINGS

An embodiment of the invention will now be described by way of example only with reference to the accompanying drawings in which:

FIG. 1 is a schematic representation of a communication system;

FIG. 2 is a schematic representation of an information frame exchanged in the communication system of FIG. 1;

FIG. 3 is a schematic representation of a frame control portion of the frame of FIG. 2;

FIG. 4 is a schematic representation of a method performed by a sender in FIG. 1;

FIG. 5 is a schematic representation of a method performed by a recipient in FIG. 1;

FIG. 6 is a schematic representation of a network protocol used in one embodiment of the communication system;

FIG. 7 is a schematic representation of an embodiment of the communication system;

FIG. 8 is a schematic representation of another embodiment of the communication system.

FIG. 9 is a schematic representation of another frame;

FIG. 10 is a schematic representation of a method performed by a sender using the frame of FIG. 9;

FIG. 11 is a schematic representation of a method performed by a recipient using the frame of FIG. 9;

FIG. 12 is a schematic representation of another communication system; and

FIG. 13 is a schematic representation of a method performed by a correspondent in FIG. 12.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

Referring to FIG. 1, a communication system 10 includes a pair of correspondents 12, 14 connected by a communication link 16. Each correspondent 12, 14 includes a respective cryptographic unit 18, 20.

Each correspondent 12, 14 can include a processor 22, 24. Each processor may be coupled to a display and to user input devices, such as a keyboard, mouse, or other suitable devices. If the display is touch sensitive, then the display itself can be employed as the user input device. A computer readable storage medium (not shown) is coupled to each processor 22, for providing instructions to the processor 22, 24 to instruct and/or configure processor 22, 24 to perform steps or algorithms related to the operation of each correspondent 12, 14, as further explained below. The computer readable medium can include hardware and/or software such as, by way of example only, magnetic disks, magnetic tape, optically readable medium such as CD ROM's, and semiconductor memory such as PCMCIA cards. In each case, the medium may take the form of a portable item such as a small disk, floppy diskette, cassette, or it may take the form of a relatively large or immobile item such as hard disk drive, solid state memory card, or RAM provided in a support

system. It should be noted that the above listed example mediums can be used either alone or in combination.

In order to transfer data between the correspondents **12**, **14**, a packet stream **30** is assembled at one of the correspondents in accordance with a defined protocol. The packet stream **30** is shown schematically in FIG. **2** and is composed of one or more frames **31**, each of which has a header **32** and data **34**. In some protocols, the packet may itself be organized as a frame with a header **32a** and the data **34a** consisting of a collection of individual frames. The header **32** is made up of a string of bits and contains control information at specified locations within the bit stream.

Included in each of the headers **34** are security control bits **33**, that include a security mode bit **35** and integrity level bits **36,37**.

In this embodiment, security mode bit **35** is used to indicate whether encryption is on or off. Integrity level bits **36** and **37** together are used to indicate which of four integrity levels, such as 0, 32, 64, or 128 bit key size is utilised. The security mode bit **35** may be used to indicate alternative modes of operation, such as, authentication and the number of bits may be increased (or decreased) to accommodate different combinations. It will be recognized that providing security bits in each frame **31** of the stream **30** allows the security level to be on a frame-by-frame basis rather than on the basis of a pair of correspondents, therefore providing greater flexibility in organizing communications.

In order to provide security, certain minimum security levels may be used. These levels should be decided upon among all of the correspondents through an agreed-upon rule. This rule may be either static or dynamic.

In operation, the correspondent **12** performs the steps shown in FIG. **4** by the numeral **100** to send information to the correspondent **14**. First, the correspondent **12** prepares data and a header at step **102**. Then it selects the security level at step **104**. The security level is determined by considering the minimum security level required by the recipient, the nature of the recipient, and the kind of data being transmitted. If the security level includes encryption, then the correspondent **12** encrypts the data at step **106**. If the security level includes authentication, then the correspondent **12** signs the data at step **108**. Then the correspondent **12** includes bits indicating the security mode and security level in the frame control at step **110**. The correspondent **12** then sends the frame to the correspondent **14** at step **112**.

j Upon receiving the frame, the correspondent **14** performs the steps shown in FIG. **5** by the numeral **120**. The correspondent **14** first receives the frame at step **122**. It then extracts the security bits at step **124**. If the mode security bits **34** indicate encryption, then the correspondent **14** decrypts the data at step **126**. If the security bits indicate authentication, then the correspondent **14** verifies the signature at step **126**. Finally, the correspondent **14** checks the security level to ensure it meets predetermined minimum requirements at step **128**. If either the encryption or authentication fails, or if the security level does not meet the minimum requirements, then the correspondent **14** rejects the message and, if the encryption and authentication do not fail, and the security level meets the minimum requirements then the message is accepted, at step **130**.

It will be recognized that providing security bits and an adjustable security level provides flexibility in protecting each frame of the communication. It is therefore possible for the sender to decide which frames should be encrypted but not authenticated. Since authentication typically increases

the length of a message, this provides a savings in constrained environments when bandwidth is at a premium.

In a further embodiment, the correspondent **12** wishes to send the same message to multiple recipients **14** with varying minimum security requirements. In this case, the correspondent **12** chooses a security level high enough to meet all of the requirements. The correspondent **12** then proceeds as in FIG. **4** to assemble and send a message with the security level. The message will be accepted by each recipient since it meets each of their minimum requirements. It will be recognized that this embodiment provides greater efficiency than separately dealing with each recipient's requirements.

In another embodiment, a different number of security bits are used. The actual number of bits is not limited to any one value, but rather may be predetermined for any given application. The security bits should indicate the algorithm parameters. They may be used to determine the length of a key as 40 bits or 128 bits, the version of a key to be used, or any other parameters of the encryption system.

It will be recognized that in the above embodiments, a network stack may be used to organize communications between the correspondents. Referring therefore to FIG. **6**, the a network stack of correspondent A is shown by the numeral **130**. A network stack of correspondent B is shown by the numeral **140**. The network stacks are organized into layers and have similar structures. The network stack **130** includes an application layer (APL) **132**, a network layer (NWK) **134**, a message authentication layer (MAC) **136**, and a physical layer (PHY) **138**. The network stack **140** includes similar components with similar numbering.

The sender determines how he wants to protect payload (and where to protect it, i.e., which layer). For the APL layer, security should be transparent; its role is limited to indicating at which level it wants to protect data (i.e., security services: none, confidentiality, data authenticity, or both). The actual cryptographic processing then is delegated to lower layers.

The recipient determines whether or not to accept protected payload, based on the received frame and locally maintained status information. The outcome of the cryptographic processing (done at the same layer as that of the sender), including info on the apparently offered protection level, is passed to the application layer, who determines whether the offered protection level was adequate. The recipient may acknowledge proper receipt of the frame to the original sender, based on this 'adequacy test'.

The acknowledgement (ACK), if present, is then passed back to the sender and passed up to the appropriate level (if protected message sent at APL layer, then ACK should also arrive back at that level; similar for lower layers of course).

The sender A determines that it wants to protect payload m using the protection level indicated by SEC (taking into account its own security needs and, possibly, those of its intended recipient(s)). The payload m and desired protection level SEC is then passed to a lower layer (e.g., the MAC layer, as in the diagram) which takes care of the actual cryptographic processing. (This message passing could include additional status information that aids in the processing of the frame, such as the intended recipient(s), fragmentation info, etc. Note that the delegation of the cryptographic processing to a lower layer is only a conceptual step if cryptographic processing takes place at the same layer at which the payload m originates.)

Cryptographic processing involves protecting the payload m and, possibly, associated information such as frame headers, using the cryptographic process indicated by the

desired protection level SEC . The key used to protect this information is derived from shared keying material maintained between the sender and the intended recipient(s). After cryptographic processing, the protected frame, indicated by $[m]K$, SEC in FIG. 6, is communicated to the intended recipient(s) B.

The intended recipient (s) retrieves the payload m' from the received protected frame, using the cryptographic process indicated by the observed protection level SEC' , using a key that is derived from shared keying material maintained between the sender and the recipient(s) in question. The retrieved payload m' and the observed protection level SEC' is passed to the same level at which the payload was originated by the sender, where the adequacy of the observed protection level is determined. The observed protection level SEC' is deemed sufficient, if it meets or exceeds the expected protection level SEC_0 , where the parameter SEC_0 might be a fixed pre-negotiated protection level that does or does not depend on the retrieved payload m' in question. (Defining SEC_0 in a message-dependent way would allow fine-grained access control policies, but generally involves increased storage and processing requirements.)

The above approach works in contexts where expected and observed protection levels can be compared, e.g., where the set of protection levels is a partial ordering or where a membership test is performed (one of a set of protection levels). One example is the context where protection involves a combination of encryption and/or authentication, with as ordering the Cartesian product of the natural ordering for encryption (encryption OFF < Encryption ON) and the natural ordering of authentication (ordered according to increasing length of data authenticity field). Moreover, if the set of protection levels has a maximum element, then the sender can use this maximum protection level to ensure that (unaltered) messages always pass the adequacy test. In another example, the observed protection level is compared to SEC_0 , where SEC_0 is a set of protection levels rather than only a minimum security level. In this way, if $SEC_0 = \{\text{None}, \text{Auth-32}, \text{Auth-64}, \text{Auth-128}\}$ and $SEC = \text{Auth-32}$, then the adequacy test would pass, whereas if SEC_0 is the same as above and $SEC = \text{Auth-32} + \text{Confidentiality}$ (e.g. encryption), then the adequacy test would fail.

In the above embodiments, each sender pre-negotiates the minimum expected protection level SEC_0 with each intended recipient. Thus, the approach might not be as adaptive as desirable for some applications and may involve additional protocol overhead at every change of the SEC_0 parameter. These disadvantages can be overcome by using the acknowledgement (ACK) mechanism from recipient(s) to sender as a feedback channel for passing the SEC_0 info. This is performed by incorporating in each acknowledgement message an indication as to the expected protection level. This information can then be collated by the original sender to update the minimum protection level expected by its recipient(s), whether or not this is message-dependent or not.

In a further embodiment, a method of synchronizing security levels is shown. Referring to FIG. 7, another embodiment of the communication system is shown generally by the numeral 160. The system includes a sender A 162 and recipients 168 in a group labelled G. The sender A includes parameters SEC_A 164 and SEC_G 166.

Sender A wants to securely communicate a message m to a group G of devices. The sender A has access to the two parameters, e.g., (1) The minimum level SEC_A at which it would like to protect this message (in general, SEC_A might depend on the group it sends information to and the message

itself, so proper notation would be $SEC_A(m, G)$); (2) The minimum protection level SEC_G that the group G of recipients expects (again, the proper notation would be $SEC_G(m, A)$ if this level would depend on the sender and the message itself as well). Here, the minimum expectation level of a group is the maximum over all group members of the minimum expectation level for each group member.

Initialization:

Sender A assumes that each parameter SEC_G is set to the maximum protection level (for each group G it securely communicates with).

Operational Usage:

Sender A determines the minimum protection level SEC_A at which it wants to protect the message m . The actual protection level SEC applied to the message m meets both its own adequacy test (i.e., $SEC \geq SEC_A$) and the minimum expected level by the group G (i.e., $SEC \geq SEC_G$).

Each recipient B that is in the group G of recipients (i.e., $B \in G$) indicates in its secure acknowledgement message the minimum expected protection level (for sender A and message m) at that particular moment of time.

A updates the parameter SEC_G such that it is consistent with all the minimum protection levels indicated in each of the acknowledgement messages it received back (i.e., $SEC_G \geq SEC_B$ for all responding devices B).

Note that the procedure described above sends messages with a protection level that satisfies both the needs of the sender and expectations of recipient(s) and is adaptable to changes herein over time. Alternatively, the sender might only take its own protection needs into account, at the cost of potentially sending messages that will be rejected by one or more recipients due to insufficient—since less than expected—protection level.

The procedure described above can be generalized towards a general self-synchronization procedure for status information among devices in any network topology, where the feedback info on status information may be partially processed along the feedback path from recipient(s) towards sender already, rather than at the sender itself only (in the example above, this graph is a tree with root A and leaves the recipient(s) and the synchronization involves a specific security parameter).

As seen in FIG. 8, A sends a payload secured at protection level SEC to a group of devices consisting of B1-B4. The recipients B1-B4 provide feedback to the sender A on the expected protection level (indicated in the diagram as the integers 1, 3, 2, 5, where these integers are numbered in order of increasing protection level). The feedback is communicated back to A via intermediate nodes C1 and C2, who collect the respective feedbacks of devices in their respective groups G1 and G2 and process this, before returning a condensed acknowledge message representing both groups to sender A. The condensed feedbacks provided by these intermediate devices provides A with the same information on the minimum protection level that satisfies the expectations of all recipients as would have been the case if this information would have been forwarded to A without intermediate processing. (Here, we assume that the intermediate devices do not cheat in their calculations)

In another embodiment, each frame in the communication is structured as shown in FIG. 9 and is generally denoted by numeral 170. The frame 170 generally comprises a header 172, a payload 174 and a footer 176. The footer 176 typically comprises one or more bits that represent an error code. The payload 174 includes the data which is to be sent in that particular frame 170, e.g. a message.

An exemplary header **172a** is also shown in greater detail in FIG. 9. The header **172a** includes a key identifier **178**, a representation of a key **180**, a frame type **182**, a security level **184** (as before) and an indication of the originator **186** of the message, e.g. the sender **12**.

Each portion of header **172a** contains one or more bits that represents a certain attribute of the transmission or includes a piece of information. The key identifier **178** and the representation of the key **180** are typically used to determine not only what key is to be used but also how the key is to be used, e.g. for broadcast or unicast communications.

The frame type **182** provides an indication as to what type of transmission is being sent in that particular frame **172a**. Typical frame types **182** include data, command, acknowledgement and beacon frames. Data-type frames transmit data, command-type frames transmit commands, acknowledgement-type frames transmit information back to a sender, e.g., an acknowledgement from the recipient that a frame has been properly received, and beacon frames are typically used to divide a transmission into time intervals.

In order to provide security, in addition to providing a minimum security level for the recipient **14**, the sender **12** includes the frame type **182** in the header **172a**. The frame type **182** is used by the recipient **14** to perform a policy check to determine if the security level, key, key usage, etc. are appropriate for the type of frame being transmitted. For example, inadequate security for a frame type that should normally include high security would be rejected.

In operation, the sender **12** performs the steps shown in FIG. 10 by the numeral **200** to send information to the recipient **14**. First, the sender **12** prepares the frame at step **202** according to steps **102-110** discussed above. It will be appreciated that these steps would also include preparation of the header **172a** to include the representation of the bits shown in FIG. 9. At step **204**, the sender **12** determines the frame type **182** and includes one or more bits into the header **172a** to indicate the frame type **182**. The sender **12** then sends the frame **170** to the recipient **14** at step **206**.

Upon receiving the frame **170**, the recipient **14** performs the steps shown in FIG. 11 by the numeral **208**. The recipient **14** first receives the frame at step **210** and then performs the steps **124-126** discussed above at step **212**. The recipient **14** then extracts the frame type **182** from the header **172a** at step **214**. The frame type **182** is then correlated to a policy in order to perform a policy check at step **216**. In particular, a look-up-table is accessed by the recipient that indicates one or more policy for each frame type. The recipient **14** then determines if the policy is met at step **218** and either rejects or accepts the frame **170** at step **220** based on whether or not the policy is met.

The policy check includes a correlation of the frame type **182** to some other data, preferably something included in the frame. For example, the policy may include certain correlations between key types and frame types such that based on the representation of the key the frame is accepted or rejected depending on whether or not the key is acceptable for use with the particular frame type **182**. In the result, a certain type of key (or key usage) is required in order for the policy to be met. If the key is not of the correct type, then the frame **170** is not accepted by the recipient **14**. If a single header **32a** is used for multiple frames **34a** as shown in FIG. 2 then the policy will also apply to the remaining frames in the message.

In another example, the policy is set based on the security level **184** that is included in the frame **170**, e.g. minimum security level SEC_0 discussed above. The frame **170** includes a certain minimum security level that has been

included at the time when the header **172** is prepared by the sender **12**, and this minimum security level is correlated to the particular frame type **162**. If the security level **184** is suitable for the frame type **162** then the frame **170** is passed by the recipient at step **220** and if not it is rejected. It will be appreciated that the policy can be adapted to correlate any suitable information included in the frame with the frame type **182**.

The above principles enable security checks to be adapted to various messages, frame types etc. in order to protect against combinations of security features that are more prone to an attack. For example, a policy can cause a recipient to reject a frame for using no encryption and only authentication, when that frame type is particularly vulnerable to an attack when encryption is not used.

In general there are three security level checks that possess different levels of granularity. The first is where SEC_0 is message independent. In this case, the minimum level of security is set once, and only one value needs to be stored locally for performing a policy check. However, where SEC_0 is message independent, a minimum granularity is provided since there is only one minimum security level for all messages and message types.

The second is where SEC_0 is completely message-dependent. In this case, a high level of granularity is provided since each message has its own minimum security level. However, this requires an enumeration of all messages and corresponding minimum security levels to be stored locally in a table.

The third is where SEC_0 is partially message dependent, namely, as discussed making reference to FIGS. 9-11, messages are divided into different types (e.g. by frame type) and, a minimum security level is allocated to each message type. This case balances the competing space requirements and granularity of performing a policy check based on the minimum security level. Typically, the number of messages/frame types is significantly less than the number of messages/frame types and thus more feasible to implement in a table.

In another embodiment shown in FIG. 12 a network N comprises one or more correspondents (e.g. A, B) communicating through a central correspondent C. Correspondent A communicates over the network N by transmitting frames **150** to the central correspondent C using, e.g., any of the principles described above. When correspondent A first wishes to engage the network N, they do not have a key and thus cannot be authenticated to communicate in the network N. The general steps for an initialization procedure are shown in FIG. 13. The correspondent C first obtains an indication that A wishes to join the network N at step **224**. This indication can be provided through a suitable registration procedure. Correspondent C then includes A in a table that indicates its status, and sets the status for correspondent A to "Exempt" at step **226**. The exempt status takes into account that an initialization procedure is required so that correspondent A can communicate unsecurely until it has been initialized in the network N.

At step **228**, correspondent A sends a frame to central correspondent C. Correspondent C checks the table at step **230**. In this first communication, the status of correspondent A is exempt and a key exchange or other initialization procedure is carried out at step **232** and the status of correspondent A is then changed to "not exempt" (or an exempt indicator is removed, set to zero etc.) at step **234**. Correspondent A then sends frames to correspondent C subject to normal security rules. At step **230** the status for correspondent A would thereafter be determined as not

exempt and the regular security rules are applied at step 236, e.g. by checking the security level, frame type etc. It can be appreciated that A could also exempt C such that the roles are reversed and A is allowing C to communicate therewith (e.g. where A is part of another network).

In an example implementation of the network N shown in FIG. 12, the above minimum security level test takes into account the frame 150 and also the originator 186. In this case, the sender is correspondent A and the recipient is correspondent B. A check for the minimum security level would thus be $SEC \geq SEC_B(m, A)$. If the minimum security level is independent of originator A, this comes down to check $SEC \geq SEC_B(m)$, as discussed before. The same storage considerations as with original security level test would then be used (case 1).

If the minimum security level is completely dependent on the originator A, a minimum security level table is enumerated (dependent on frame m, frame type of m, or message dependent—as discussed before), but now for each originator (case 2). If the minimum security level is independent of originator A, except when originator is in an explicitly enumerated set of exempt devices, e.g. denoted by ExemptSet in the table, a single minimum security level table is implemented for devices outside the ExemptSet (potentially depending on frame type, etc.) and, additionally, a minimum security level table for each individual member of ExemptSet is implemented (case 3). Thus, if a correspondent (and device associated therewith) is part of the ExemptSet table, case 2 is utilized and, if no device is in the ExemptSet table, case 1 is utilized.

Case 3 can be made more implementation-friendly if correspondents in the ExemptSet table, a minimum security level table that is independent of the particular device in the ExemptSet is used. This requires that one security level table is implemented for devices that are not in the ExemptSet table and one table is implemented for devices that are in the ExemptSet table (case 4).

A further optimization of case 4 is where, for all devices in the ExemptSet table, the minimum security level—which is potentially message or message type dependent (as discussed before)—is either set to the minimum security level that holds for all devices that are outside ExemptSet or is set to a prespecified value that may hold for all devices inside ExemptSet. Since this would lead to only two choices (e.g. per frame, frame type, overall), this can be indicated using a Boolean parameter.

In summary:

$SEC \geq SEC_B(m, A)$, where

$SEC_B(m, A) = SEC_B(m)$ if A is not a member of ExemptSet.

$SEC_B(m, A) = SEC_B(m)$ if A is a member of ExemptSet and Override parameter OverrideSEC(m) for message m is set to FALSE.

$SEC_B(m, A) = \text{ExemptSEC}_B(m)$ if A is a member of ExemptSet and Override parameter OverrideSEC(m) for message m is set to TRUE.

In general, the most practical scenario is where ExemptSEC_B(m) is set to 'no security'.

It is noted that one scenario allows devices that do not have a key yet (e.g., because these just joined the network and still have to set up a key, e.g., via key agreement or key transport protocol or PIN or any other mechanism) to "by-pass" the minimum security level check (i.e., the security check always succeeds), if these have been labeled by recipient B as belonging to ExemptSet (and ExemptSEC_B(m) is set to 'no security').

The by-passing of minimum security level checks may depend on the message m received, the frame type of message in (which is visible to the recipient if the frame type of m is included in the transmitted frame—normally frame types and other frame control information is not encrypted), or a parameter that can be set via the Override parameter OverrideSEC(m).

It is also noted that operations on the set ExemptSet by the recipient effectively govern the operation of the minimum security level check (inclusion of a device in that set may allow by-passing or lowered security requirements, while exclusion of a device from that set restores the ordinary minimum security level check and make it applicable (possibly again) to the originating device in question).

Thus, the above allows a flexible mechanism to take into account transitional behaviour of a correspondent (and their device) during the system's lifetime, and facilitates the transgression of a device from some initial stage where it does not yet have a key, to the stage where it has established a key and can be enforced to adhere to normal strict minimum security level policies.

The override parameter OverrideSEC(m) allows fine-tuning of "by-passing" the minimum security level check and make this dependent on the message m received (or message type—obviously one can make granularity as fine-grained as possible, at expense of table implementation cost). As an example, in the scenario where a device joins a network and still has to set up a key, one could set the Override parameter to TRUE only for those messages or message types minimally required for the originating device A to set up a key with the recipient device B (or with some other device T in the network that could notify B once the key has been established), thus restricting the permissible behavior of device A, but not ruling out all behaviors. This can also be used for any other initialization procedure or set-up procedure and should not be limited to key set up.

Again, operations on the Override parameter Override(m) by the recipient B allow for a very flexible and low-cost fine-tuning of security control policies. As an example, by setting all Override parameters to FALSE, one effectively closes down all network operations to devices that do not have a key (since all cryptographically unsecured messages to recipient B will ultimately be rejected)—the so-called stealth mode—while setting all Override parameters to TRUE allows unlimited flows of unsecured information to device B, since this may result in the minimum security level test to be effectively by-passed.

It will be recognized that the security rules can be adapted to provide flexibility not only on a frame-by-frame basis but also based on the frame type such that a policy check can determine if certain security rules or key types can be used with a particular frame type.

Although the invention has been described with reference to certain specific embodiments, various modifications thereof will be apparent to those skilled in the art without departing from the spirit and scope of the invention as outlined in the claims appended hereto.

The embodiments of the invention in which an exclusive property or privilege is claimed are defined as follows:

1. A method, comprising:

receiving, by a first device, a frame from a second device, wherein the frame comprises a header and data, and the header indicates a first minimum security level, and wherein the header comprises a frame type, a key identifier of a key and a key representation of the key, wherein the key identifier and the key representation indicate whether the key is used for broadcast commu-

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nication or unicast communication, wherein the frame type corresponds to a particular security level; determining, by the first device, whether the frame is acceptable based on the particular security level corresponding to the frame type included in the header, the first minimum security level indicated by the header, the key identifier included in the header and the key representation included in the header, wherein the determining whether the frame is acceptable comprises comparing the first minimum security level indicated by the header with the particular security level corresponding to the frame type in the header.

2. The method of claim 1, wherein the first minimum security level comprises an authentication level.

3. The method of claim 1, wherein the first minimum security level comprises an encryption level.

4. A first device, comprising:
 at least one hardware processor; and
 a non-transitory computer-readable storage medium coupled to the at least one hardware processor and storing programming instructions for execution by the at least one hardware processor, wherein the programming instructions, when executed, cause the first device to perform operations comprising:
 receiving a frame from a second device, wherein the frame comprises a header and data, and the header indicates a first minimum security level, and wherein the header comprises a frame type, a key identifier of a key and a key representation of the key, wherein the key identifier and the key representation indicate whether the key is used for broadcast communication or unicast communication, wherein the frame type corresponds to a particular security level; and
 determining whether the frame is acceptable based on the particular security level corresponding to the frame type included in the header, the first minimum security level indicated by the header, the key identifier included in the header and the key representation

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included in the header, wherein the determining whether the frame is acceptable comprises comparing the first minimum security level indicated by the header with the particular security level corresponding to the frame type in the header.

5. The first device of claim 4, wherein the first minimum security level comprises an authentication level.

6. The first device of claim 4, wherein the first minimum security level comprises an encryption level.

7. A non-transitory computer-readable medium storing instructions which, when executed, cause a first device to perform operations comprising:
 receiving a frame from a second device, wherein the frame comprises a header and data, and the header indicates a first minimum security level, and wherein the header comprises a frame type, a key identifier of a key and a key representation of the key, wherein the key identifier and the key representation indicate whether the key is used for broadcast communication or unicast communication, wherein the frame type corresponds to a particular security level; and
 determining whether the frame is acceptable based on the particular security level corresponding to the frame type included in the header, the first minimum security level indicated by the header, the key identifier included in the header and the key representation included in the header, wherein the determining whether the frame is acceptable comprises comparing the first minimum security level indicated by the header with the particular security level corresponding to the frame type in the header.

8. The non-transitory computer-readable medium of claim 7, wherein the first minimum security level comprises an authentication level.

9. The non-transitory computer-readable medium of claim 7, wherein the first minimum security level comprises an encryption level.

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